

VAHA — APPENDIX

Supporting Material for the TKS Activate Earthshot Challenge 2026

Vaha is the missing exchange layer for Bengaluru’s treated water. Apartment complexes already generate treated wastewater through on-site STPs. Construction sites nearby already buy tanker water for non-potable use. What is missing is the trusted system that turns those two facts into a repeatable market. Vaha supplies that system through matching, verification, documentation, and route coordination.

1. The Market Gap

Bengaluru’s water problem is structural, not seasonal. WELL Labs’ urban water balance work estimates citywide demand at roughly **2,632 MLD**, against Cauvery-derived piped supply of roughly **1,460–1,470 MLD**.¹ Groundwater fills part of that gap, but the 2024 East Bengaluru crisis showed how fragile that dependence has become.

At the same time, decentralised treatment capacity already exists at meaningful scale. WELL Labs’ Bengaluru reuse strategy states that the city has **3,500+ decentralised sewage treatment plants** and that current systems could support reuse of up to **615 MLD** of treated wastewater.² Public reporting and stakeholder discourse consistently frame the apartment-surplus opportunity around roughly **300 MLD**. That is the wedge Vaha is designed around, because it is large enough to matter and narrow enough to model clearly.

This is what makes the opportunity so unusual. Bengaluru is not waiting for treatment technology to be invented. The city already treats water every day. The market failure is that treated water and nearby non-potable demand remain disconnected.

2. Why Bengaluru, and Why Now

Apartment STPs were built into Bengaluru’s residential growth pattern long before a resale market existed. Recent policy action changed that. In 2024, BWSSB enabled apartment communities to commercially sell up to **50%** of treated STP water for non-potable applications such as construction, while excluding food and medicine industries.³⁴

¹https://welllabs.org/wp-content/uploads/2023/10/WELL-Labs_Bengaluru-Urban-Water-Balance-Report.pdf

²<https://welllabs.org/strategies-safe-reliable-decentralised-water-reuse-bengaluru/>

³<https://timesofindia.indiatimes.com/city/bengaluru/apartments-sign-pact-withbwssb-to-sell-stp-water/articleshow/109259296.cms>

⁴<https://timesofindia.indiatimes.com/city/bengaluru/kspcb-warns-no-oc-if-treated-water-not-used-for-construction/articleshow/110545705.cms>

That timing matters. For almost a decade, the infrastructure existed without a clean commercial route. Today, the infrastructure exists and the legal opening exists. The operating layer still does not.

Construction is the right first buyer because it already purchases non-potable water, already receives tanker deliveries, and already has clear use cases: curing, dust suppression, wheel wash, and site cleaning. It also fits the policy direction. KSPCB has pushed builders to use treated water for construction, and IISc researchers were cited publicly saying STP-treated water is suitable for such non-potable uses when norms are met.⁵

3. Why This Has Not Been Solved Already

The problem is not treatment. It is **trust, transport, and coordination**.

The trust problem is simple: construction buyers do not just ask whether water is cheaper. They ask whether they can rely on what is being delivered, whether it matches the intended use, and who is accountable if something goes wrong.

The transport problem is equally practical: even if a source and buyer are close to one another, there is no default, trusted, repeatable route that moves treated water from apartment STPs to site sumps.

The coordination problem is what ties the first two together. Supply exists. Demand exists. Policy exists. But there is no standardized layer handling quality documentation, dispatch, chain-of-custody, and payment flow.

This is why the BWSSB and BAF initiative matters so much. It proves the city recognizes the market. But later reporting makes clear that the effort remains limited because the facilitator layer is still missing.⁶

4. What Vaha Actually Does

Vaha turns surplus STP water into a verified local supply loop in three steps:

- **Match:** identify apartment STP surplus and pair it with nearby construction demand.
- **Verify:** issue a Water Acceptance Pack before dispatch so the buyer can review the source, intended use, and quality documentation.
- **Deliver:** use existing tanker operators and existing site workflows to complete the route.

The point is not to replace the underlying infrastructure. STPs already exist. Tankers already exist. Buyers already exist. Vaha creates the trusted operating protocol that lets those systems transact cleanly and repeatedly.

⁵<https://timesofindia.indiatimes.com/city/bengaluru/kspcb-warns-no-oc-if-treated-water-not-used-for-construction-articleshow/110545705.cms>

⁶<https://timesofindia.indiatimes.com/city/bengaluru/bengaluru-water-supply-and-sewerage-boards-push-for-treated-water-articleshow/130360512.cms>

5. The Water Acceptance Pack

The Water Acceptance Pack is one of the most important components in the model because it translates the abstract idea of “trust” into an operating workflow.

At minimum, the pack includes:

- source STP name and location,
- buyer / site name,
- intended use classification,
- latest lab report date,
- tested parameters,
- tanker ID,
- pickup timestamp,
- delivery timestamp,
- buyer signoff,
- complaint and escalation protocol.

This is the step that turns “alternative water” into something a site manager can actually evaluate and accept. It is also the beginning of a chain-of-custody record, which is critical if Vaha is going to be taken seriously as a real operating company rather than as a loose broker.

6. Unit Economics

The route model used in the deck is deliberately simple and consistent. Both the problem slide and the economics slide now use the same unit: a **12,000L load**.

The comparison on the problem side is:

- treated water equivalent: **Rs 120–960 / 12,000L load**
- crisis tanker equivalent: **up to Rs 2,400 / 12,000L load**

The working route example on the economics side is:

Component	Amount	Interpretation
Site pays per 12,000L load	Rs 1,200	materially below crisis tanker pricing
Apartment payout	Rs 120	Rs 10/kL equivalent
Tanker logistics	Rs 700–740	local short-route assumption
Vaha gross margin	Rs 340–380	gross margin before overhead

This framing is strong for three reasons. First, the buyer still saves meaningfully against the highest tanker-water benchmark used in the deck. Second, the apartment still earns money from water that would otherwise remain underused or difficult to move. Third, Vaha’s economics become strong enough to look like a business, not just a good deed.

How the numbers are calculated

The model uses four linked steps.

First, the addressable wedge is defined as **300 MLD** of apartment-surplus treated water. This number is not meant to describe every wastewater flow in Bengaluru. It is the focused surplus market that sits closest to the apartment-to-construction exchange Vaha is targeting.

Second, the model converts that volume into route-sized units. A single load is defined as **12,000L**, which is equivalent to **12 kL**. Since **300 MLD = 300,000,000L/day**, full capture of that wedge would equal:

$$300,000,000\text{L} / 12,000\text{L} = 25,000 \text{ loads per day}$$

Third, capture rates are applied to that route base:

- 1% capture = 250 loads/day
- 5% capture = 1,250 loads/day
- 10% capture = 2,500 loads/day
- 20% capture = 5,000 loads/day

Fourth, the route margin is applied. The midpoint route margin used in the deck is **Rs 360/load**. So, for example, the 1% case becomes:

$$250 \text{ loads/day} \times \text{Rs } 360/\text{load} = \text{Rs } 90,000/\text{day}$$

Annualized:

$$\text{Rs } 90,000/\text{day} \times 365 = \text{Rs } 3.29 \text{ Cr/year}$$

The USD conversions shown in the deck are simply secondary readability figures for judges. The operating logic is built in INR because that is how the local market actually behaves.

Where the numbers come from

The model draws from four evidence bases:

- WELL Labs’ Bengaluru reuse work for decentralised STP scale and reuse potential
- current public reporting and stakeholder discussion for the apartment-surplus wedge
- current market pricing comparisons between treated-water sale values and tanker-water prices
- a route design that keeps transport local enough for the economics to remain attractive

The treated-water side of the deck uses the known range of **Rs 10–80/kL**, which translates to **Rs 120–960 per 12,000L load**. The tanker side uses the high-end crisis benchmark of **Rs 200/kL**, which translates to **up to Rs 2,400 per 12,000L load**. The chosen Vaha route price, **Rs 1,200/load**, is intentionally positioned between those two poles. It is meaningfully better for the buyer than the highest tanker benchmark, while leaving a healthier margin for Vaha than the most conservative route model did.

Scaling the route model

The scale chart in the deck uses the following assumptions:

- apartment-surplus market: **300 MLD**
- load size: **12,000L**
- 100% capture equivalent: **25,000 loads/day**
- midpoint Vaha gross margin: **Rs 360/load**

Capture	Loads / day	Annual gross margin (INR)	Annual gross margin (USD)
1%	250	Rs 3.29 Cr	~\$353K
5%	1,250	Rs 16.43 Cr	~\$1.77M
10%	2,500	Rs 32.85 Cr	~\$3.53M
20%	5,000	Rs 65.70 Cr	~\$7.06M

These figures are shown as **gross margin before overhead**. That is the right level of confidence for an early-stage route model: strong enough to demonstrate real business potential, but honest about the fact that route economics are not the same as final company profit.

7. Competitive Positioning

Boson Whitewater is the clearest proof that treated-water demand is already real in Bengaluru. Boson’s model appears more vertically integrated around upgraded water quality, direct supply, and industrial-grade use cases.⁷⁸

⁷<https://bosonwhitewater.com/Low-TDS-Tanker-Water-Supply>

⁸<https://newskarnataka.com/bengaluru/bengaluru-turns-to-treated-wastewater-as-industrial-water-demand-surges/>
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That strengthens Vaha’s case rather than weakening it. Boson demonstrates that:

- buyers will pay for alternative water,
- quality assurance matters,
- tanker-delivered treated-water markets already exist.

Vaha’s position is different. It does not try to out-Boson Boson. It starts by solving the exchange problem: matching surplus treated water to nearby non-potable demand through a lighter, asset-efficient operating model.

One simple way to understand that difference is this: Boson behaves more like a vertically integrated delivery utility, while Vaha behaves more like a marketplace and trust layer. Boson owns more of the water-quality and supply stack. Vaha is designed to coordinate existing sources, existing carriers, and existing buyers with far less owned infrastructure.

Tankerwala-style aggregators prove that tanker coordination at city scale can work.⁹¹⁰ Vaha’s difference is focus. The first version is not broad. It is tightly scoped to one corridor, one buyer segment, and one trust workflow.

8. Validation

The validation story rests on three layers.

Operator-side validation

In a direct exchange, Vikas Brahmavar of Boson Whitewater confirmed that adoption comes down to three things: **quality, reliability, and cost**. That matters because those are the same three pillars Vaha is designed around.

Buyer-side validation

Buyer-side discovery reinforced the same point from the opposite direction: site managers do not switch just because water is cheaper. They switch when they believe the source is trustworthy and the handoff is properly documented.

Market proof

WELL Labs’ published work showed that **88%** of potential users are within **2 km** of an existing or planned DSTP, and that demand is significantly larger than decentralised supply.¹¹ Urban Acres reports that South City apartment complex already processes **9–9.5 lakh litres/day** of treated water and sells surplus water, while Times of India reported that some apartment complexes were

⁹<https://tankerwala.in/about/>

¹⁰<https://timesofindia.indiatimes.com/city/bengaluru/city-apartments-not-finding-tankers-to-sell-treated-water/articleshow/108958603.cms>

¹¹<https://welllabs.org/workshop-decentralised-wastewater-reuse/>

selling excess treated water for roughly **Rs 100–125 per tanker** even as others struggled to find transport.¹²¹³

That is the strongest possible signal for Vaha:

The transaction is already possible. It is just not yet organized into a repeatable local market.

9. Pilot Plan

The first pilot corridor is the **Bellandur → Kadubeesanahalli** stretch of Outer Ring Road. This corridor is the right launch environment because it concentrates apartment supply, active construction demand, and short route distances in the same geography.

The first 90 days are designed around one clear proof point:

One route. Thirty paid deliveries. Zero rejected loads.

That goal is intentionally narrow. It is not a vanity milestone. It is the minimum evidence required to show that Vaha is not merely an idea, but a workable operating model.

The first phase validates:

- one apartment STP source,
- one construction buyer,
- one tanker partner,
- one quality and signoff workflow.

If that loop works, the next step is not to declare total city coverage. It is to repeat the same route logic across adjacent clusters. That is how the business scales.

How the pilot would actually be validated

The pilot is not only about whether one delivery can happen. It is about whether the loop can be trusted and repeated. That means Vaha has to validate five things in sequence.

First, the source has to be real, compliant, and consistently producing surplus treated water. Second, the buyer has to already be using non-potable tanker water and have a real receiving workflow on site. Third, the tanker partner has to be able to complete the route within the economic window. Fourth, the Water Acceptance Pack has to be strong enough that the buyer is comfortable signing off before dispatch. Fifth, the payment and proof-of-delivery flow has to be clean enough that the next route is easier, not harder.

Those are the real success conditions behind the 30-delivery target. Thirty deliveries matter because they are enough to show whether the route is becoming operationally repeatable, not just technically

¹²<https://urbanacres.in/bengaluru-residences-intensify-water-management-post-summer-water-scarcity/>

¹³<https://timesofindia.indiatimes.com/city/bengaluru/city-apartments-not-finding-tankers-to-sell-treated-water/articleshow/108958603.cms>

possible.

Who Vaha needs to talk to

The first phase of market development depends on talking to the right categories of people, not just sending broad cold outreach.

On the source side, Vaha needs apartment associations, facility managers, STP operators, and federation-level housing groups that can identify where true surplus exists and where decision-making is centralized enough to move quickly.

On the buyer side, Vaha needs construction-site project managers, contractors, civil engineers, and builder-side operations staff who control procurement for non-potable site water and can clearly state what documentation they would require before switching.

On the logistics side, Vaha needs tanker operators who already serve the corridor and understand local dispatch constraints, turnaround time, and route viability.

On the validation side, Vaha benefits from continued conversations with water researchers, policy actors, and experienced reuse operators who can pressure-test assumptions before scaling.

How Vaha would run outreach

The outreach strategy should stay focused and corridor-first.

The first wave is source discovery, identifying apartments with recent compliant lab reports and real surplus volume. The second wave is buyer discovery, validating which active construction sites are already paying for tanker water in the same geography and what assurance they need to switch. The third wave is logistics discovery, confirming route costs, turnaround constraints, and reliability with tanker partners.

This sequence matters. It prevents the company from overbuilding a marketplace before it has a corridor where source, buyer, and route are all real at the same time.

10. Risk and Control

The central risk is not that Bengaluru lacks treated water. The central risk is that trust failures, route inconsistency, or weak documentation make repeatability impossible.

That is why the first version of the company is designed around control rather than breadth. The model stays disciplined by:

- staying within non-potable use cases,
- keeping route distances short,
- using documented source reports,
- requiring pre-dispatch signoff,
- treating rejected loads as a critical learning event rather than as normal friction.

This is also why Vaha is stronger as a corridor-first company than as a citywide marketplace from day one. A real exchange layer has to earn trust through repetition.

11. Team

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12. Sources

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